

ROMAIN GERARD

French National | Software Engineering Researcher
Software Engineering (M.Eng), USTC
Suzhou, Jiangsu
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EDUCATION

University of Science and Technology of China (USTC)	Hefei/Suzhou, China
<i>M.Eng. in Electronic Information - Major: Software Engineering</i>	<i>2025 - Present</i>
Coursework/Focus: Information Security, Compiler Engineering, Blockchain Technologies, Information Theory & Coding, Ubiquitous OS, HarmonyOS Development, Artificial Intelligence	
Current GPA: 3.54/4.0 (First Semester)	

University of Rouen Normandy	Rouen, France
<i>Associate + Bachelor of Science (B.Sc.) equivalent in Multimedia & Internet Technologies</i>	<i>2020 - 2024</i>
GPA: Top of Class – 16.07/20 (4.0 equivalent)	

WORK EXPERIENCE

SYNK (Internship)	Remote
<i>Software Engineering Intern</i>	<i>January 2025 – August 2025</i>
- Developed and maintained internal Python tooling; contributed to code reviews within Agile sprint cycles across engineering teams. - Built and tested secure Python components, participated in vulnerability assessments on internal tooling, and supported CI/CD pipeline maintenance via GitLab.	

PROJECT EXPERIENCE

HexStrike-AI LLM Security Research Framework Link	Suzhou, China
<i>Security Researcher & Lead Developer</i>	<i>February 2026 - Present</i>
<i>USTC Suzhou Institute – Information Security Lab Engineering Project (Credited)</i>	
<i>Supervisor: Prof. Guo Yan (郭燕)</i>	
- Extended a Python/Flask pentest framework exposing 150+ security tools via MCP, researching why LLM agents underutilize available tools and designing an improved architecture. - Ran controlled experiments comparing Claude and DeepSeek across 4 AI clients (Claude.ai, RooCode, Sire, TRAE) on PicoCTF challenges, measuring tool-selection behavior across 3 variants: baseline preference, ranking compliance, and forced compliance. - Built a 529-prompt benchmark corpus across 7 CTF categories × 3 difficulty levels × 2 LLMs × 3 clients for reproducible behavioral analysis. - Deployed an OpenAI-compatible LLM proxy to intercept and log tool call chains across clients; defined 3 core metrics: ranking compliance rate, selection frequency bias, and decision divergence rate.	

2026 Software System Security Competition (National)	Suzhou, China
<i>Security Researcher / CTF Competitor</i>	<i>March 2026</i>
- Participated in the 2026 Software System Security Competition, a national attack-defense cybersecurity competition. - Solved challenges across forensics, reverse engineering, and cryptography.	

OS Engineering Labs – Ubiquitous Computing Course	Suzhou, China
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Graduate Coursework

Dec 2025

- Implemented a custom Linux kernel syscall in C using RCU locking for deadlock-safe process/group ID traversal; validated via kernel logs and /proc filesystem.
- Cross-compiled Linux 5.4.250 for x86 and ARM64 using GCC and QEMU; built minimal initramfs with BusyBox and custom init scripts.
- Built a buddy memory allocator in C with XOR-based partner indexing, block splitting, and cascading merge on free.
- Deployed openEuler 22.03 LTS on a two-node cluster; loaded HMDFS kernel module and validated 9 distributed filesystem behaviors including cross-node sync and write conflict detection.

SEED Labs – Network Security Practicum

Suzhou, China

Security Researcher / Graduate Coursework

Oct 2025 – Nov 2025

- Performed ARP cache poisoning (request, reply, gratuitous) in Docker environments using Scapy to enable MITM attacks on Telnet and Netcat sessions with live traffic modification.
- Exploited 32/64-bit buffer overflows via shellcode injection and GDB-assisted offset calculation, achieving root privilege escalation through Set-UID programs.
- Developed Linux kernel modules using Netfilter hooks to filter traffic by protocol, IP, and port; validated enforcement dynamically against live DNS and ICMP traffic.
- Executed ICMP redirect attacks to hijack victim routing tables, followed by TCP-layer payload interception and modification using Scapy.

Stardew Dashboard (Personal Project)

Paris, France

Full-Stack AI Developer

Feb 2024 - Apr 2025

- Co-developed an intelligent dashboard analyzing Stardew Valley game saves using ReactJS and PHP, with Python/scikit-learn for in-game data categorization.
- Added automated filtering and recommendation features, enhancing functionality and user experience.

Monarchy Mayhem (Awarded)

Paris, France

Game Developer

Nov 2023 - Feb 2024

- Built a full game in Unity (C#) with a team; awarded 3rd place at Festival MMI France.
- Engineered enemy AI using state machines and pathfinding; designed adaptive difficulty mechanics based on player performance analytics.

SKILLS & LANGUAGES

Security: Nmap, Burp Suite, Wireshark, x86 Assembly, Scapy, GDB

Dev & Infra: Python, Flask, JavaScript, PHP, C, C#, Docker, Git, Linux/Bash, GitLab, CI/CD

Languages: French (Native), English (C1 - IELTS 7.5), Spanish (B1), Mandarin (Beginner - Studying)